

Swarali Kestikar

Arlington, VA

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EDUCATION

Virginia Tech

August 2024 – May 2026

Master of Science in Computer Science | *GPA: 3.88/4.00*

Arlington, VA

- **Relevant Coursework:** Web Application Development, Machine Learning with Big Data, Natural Language Processing, Cloud Computing, Software Engineering.

MKSSS's Cummins College of Engineering for Women

August 2018 – July 2021

Bachelor of Technology in Computer Engineering

Pune, India

TECHNICAL SKILLS

Languages: Java, Python, Rust, SQL, JavaScript (ES6+), TypeScript, HTML/CSS

Frameworks & Web: React.js, Next.js, Node.js, Express.js, Spring Boot, PyTorch, Transformers, REST

Cloud & Infrastructure: AWS (EC2, Lambda, S3, API Gateway), Microservices, Azure, Docker, CI/CD (Jenkins, GitHub Actions)

Databases & Tools: MySQL, PostgreSQL, Oracle Database, MongoDB, Git, Bitbucket, Ansible, JIRA, Tableau

WORK EXPERIENCE

Amazon

May 2025 – August 2025

SDE Summer 2025 Intern

Seattle, WA

- Reduced package processing execution time by **90% (76 to 8 minutes)** by optimizing the data pipeline through **Octocrab** and **Rayon** for **high-concurrency parallelization in Rust**.
- Strengthened software supply chain security for **100+ packages** by architecting an **automated Rust-based freshness tracking tool** to monitor upstream repository deltas.
- Mitigated **enterprise-scale dependency risks** by engineering a **configurable staleness detection engine** that systematically identifies and prioritizes security vulnerabilities.

Bank of America

June 2021 – August 2024

Officer; Software Engineer

Mumbai, India

- Boosted transaction efficiency by **50%** by engineering high-performance financial applications leveraging **Python-based Quartz frameworks** and **Oracle Database clusters**.
- Led a large-scale **process re-engineering initiative**, restructuring end-to-end service processes across **7 upstream systems** and **2 downstream teams**, reducing paperwork by **70%** and enhancing cross-team collaboration.
- Eliminated **40%** of manual operational overhead by **automating mission-critical workflows** and documentation generation using **Python and Jenkins CI/CD** scripting.
- Increased production environment stability by **45%** through the design and implementation of **modular unit, integration, and performance regression testing suites**.
- Accelerated developer onboarding velocity by **25%** by implementing **automated technical documentation tools** and optimizing cross-functional knowledge transfer protocols.

TECHNICAL PROJECTS

Adversarial Training for NLP

Python, PyTorch, Transformers

- Reduced wall-clock training time by **5x** by developing a **Span Level Reconstruction (SLR)** objective that captures multi-token semantic dependencies.
- Maintained **94.45% clean accuracy** while significantly lowering computational overhead through **advanced span-level reconstruction** over baseline token methods.
- Stabilized performance within the **first epoch** by implementing **selective masking** of multi-token informative subsequences to enhance model semantic fidelity.
- Verified cross-architecture robustness transfer by architecting **joint training regimes across RoBERTa and DeBERTa** to enforce structural regularization.

Enterprise Bookstore Platform

React, TypeScript, Node.js, Docker

- Attained **100% build success rates** and dev-to-prod parity by **containerizing the microservices stack using Docker** and automating CI/CD pipelines via GitHub Actions.
- Optimized data retrieval latency by implementing a **persistent MySQL database** with **advanced indexing** to handle high-volume transaction queries.
- Fortified **transaction integrity** and system reliability by developing a full-stack e-commerce engine with **modular RESTful API integration** and TypeScript type-safety.

AWARDS & HONORS

Global Recognition Awards: Received dual Gold and Silver Awards for excellence in quality assurance and production resilience at Bank of America.